

Problem Statement

Evaluate the following limit from the file dailyintegral-limit-limit-1x1-2026-07-15.jpg:

$$\lim_{n \rightarrow \infty} \left(\frac{\cosh\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right)} \right)^{n^2}$$

Detailed Solution

This limit evaluates an indeterminate form of type 1^∞ . We can solve it by transforming the expression using the natural logarithm and applying L'Hôpital's rule. Let's break this down step-by-step.

Step 1: Express the limit using the natural logarithm

Let the value of the limit be L :

$$L = \lim_{n \rightarrow \infty} \left(\frac{\cosh\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right)} \right)^{n^2}$$

Taking the natural logarithm of both sides allows us to bring down the exponent:

$$\ln L = \ln \left[\lim_{n \rightarrow \infty} \left(\frac{\cosh\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right)} \right)^{n^2} \right]$$

Since the natural logarithm is a continuous function, we can move it inside the limit:

$$\ln L = \lim_{n \rightarrow \infty} n^2 \ln \left(\frac{\cosh\left(\frac{\pi}{n}\right)}{\cos\left(\frac{\pi}{n}\right)} \right)$$

Step 2: Change variables to simplify the limit

To make the evaluation easier, let's introduce a substitution. Let $x = \frac{\pi}{n}$. As $n \rightarrow \infty$, the new variable $x \rightarrow 0^+$. Rearranging for n , we have $n = \frac{\pi}{x}$, which means $n^2 = \frac{\pi^2}{x^2}$.

Substituting this into our limit gives:

$$\ln L = \lim_{x \rightarrow 0^+} \frac{\pi^2}{x^2} \ln \left(\frac{\cosh x}{\cos x} \right)$$

Using the quotient property of logarithms, $\ln\left(\frac{a}{b}\right) = \ln a - \ln b$, we can rewrite the expression as:

$$\ln L = \pi^2 \lim_{x \rightarrow 0^+} \frac{\ln(\cosh x) - \ln(\cos x)}{x^2}$$

Step 3: Evaluate the limit using L'Hôpital's Rule

As $x \rightarrow 0$, both the numerator $\ln(\cosh 0) - \ln(\cos 0) = \ln(1) - \ln(1) = 0$ and the denominator $0^2 = 0$. This is a $\frac{0}{0}$ indeterminate form, meaning we can apply L'Hôpital's Rule by differentiating the numerator and the denominator with respect to x .

The derivative of the numerator is:

$$\frac{d}{dx}[\ln(\cosh x) - \ln(\cos x)] = \frac{\sinh x}{\cosh x} - \frac{-\sin x}{\cos x} = \tanh x + \tan x$$

The derivative of the denominator is:

$$\frac{d}{dx}[x^2] = 2x$$

Applying L'Hôpital's Rule, the limit becomes:

$$\ln L = \pi^2 \lim_{x \rightarrow 0^+} \frac{\tanh x + \tan x}{2x}$$

Checking the limit again as $x \rightarrow 0$, we get $\frac{0+0}{0}$, which is still a $\frac{0}{0}$ indeterminate form. We must apply L'Hôpital's Rule a second time.

The derivative of the new numerator is:

$$\frac{d}{dx}[\tanh x + \tan x] = \operatorname{sech}^2 x + \sec^2 x$$

The derivative of the new denominator is:

$$\frac{d}{dx}[2x] = 2$$

Now, evaluate the limit as $x \rightarrow 0$:

$$\ln L = \pi^2 \lim_{x \rightarrow 0^+} \frac{\operatorname{sech}^2 x + \sec^2 x}{2}$$

Since $\operatorname{sech}(0) = \frac{1}{\cosh(0)} = 1$ and $\sec(0) = \frac{1}{\cos(0)} = 1$, we can substitute $x = 0$ directly:

$$\ln L = \pi^2 \left(\frac{1^2 + 1^2}{2} \right) = \pi^2 \left(\frac{2}{2} \right) = \pi^2$$

Step 4: Compute the final limit

We have successfully found that $\ln L = \pi^2$. To find our original limit L , we exponentiate both sides:

$$L = e^{\pi^2}$$

Final Answer:

$$\mathbf{e^{\pi^2}}$$